Discrete Structures

Part 1: Sets, Sequences, Integers, Matrices.

Ivan Bliznets

University of Groningen

Overview

1. Intro
2. Sets and Subsets
3. Operations on Sets
4. Sequences
5. String and Languages
6. Integers
7. Matrices
8. Mathematical Structures

Outline

1 Intro

2 Sets and Subsets

3 Operations on Sets

4 Sequences

5 String and Languages

6 Integers

7 Matrices

8 Mathematical Structures

Learning Goals of Part 1

Book chapter: **1. Fundamentals.**

Mathematical objects: sets, sequences, integers, and matrices

After completion of this module, you are able to

- specify and reason about sets, sequences, integers, and matrices.
- apply basic operations on sets, sequences, integers and matrices.
- give simple mathematical proofs of basic facts about sets, sequences, integers and matrices.
- apply the concept of mathematical structure.

Recap of set basic theory, introduce set operations.

Note: Part 1 topics do not have a strong connection to each other. They form the basis for later chapters.

The Role of Logic

In this course:

- Correct notation is important, but we will not be too strict about difference between meta-language and formal language.
- We will not give/require formal (Fitch-style) proofs.
- We will give/require informal, yet rigorous mathematical proofs.

Outline

1 Intro

2 Sets and Subsets

3 Operations on Sets

4 Sequences

5 String and Languages

6 Integers

7 Matrices

8 Mathematical Structures

Sets: Unordered Collections of Distinct Objects

A **set** is an unordered collection of distinct objects called its **elements** (or **members**).
Notation:
$x \in A$: *"x is an element of the set A"*.
$x \notin A$: *"x is not an element of the set A"*.



- Examples:
  - the set of all students in this course
  - the set of all integers
  - $V = \{\text{horse, mars, formula, cylinder, teapot}\}$

Sets: Unordered Collections of Distinct Objects

A **set** is an unordered collection of distinct objects called its **elements** (or **members**).
Notation:
$x \in A$: *"x is an element of the set A"*.
$x \notin A$: *"x is not an element of the set A"*.



- Examples:
  - the set of all students in this course
  - the set of all integers
  - $V = \{\text{horse, mars, formula, cylinder, teapot}\}$
- Each member of a set occurs only **once** and a set has **no ordering**. For example, $\{1, 1, 3, 2\}$, $\{3, 1, 3, 2, 1\}$ and $\{1, 2, 3\}$ all denote the same set. We have: $1 \in \{1, 2, 3\}$, but $0 \notin \{1, 2, 3\}$.

Sets of Numbers

$$
\begin{aligned}
\mathbb{Z} &= \{x \mid x \text{ is an integer}\} \\
&= \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\} \\
\mathbb{N} &= \{x \mid x \text{ is a natural number}\} \\
&= \{0, 1, 2, 3, \ldots\} \\
\mathbb{Z}^+ &= \mathbb{N}^+ = \{x \mid x \text{ is a positive integer}\} \\
&= \{1, 2, 3, \ldots\} \\
\mathbb{Q} &= \{x \mid x \text{ is a rational number}\} \\
&= \left\{\frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0\right\} = \left\{\frac{p}{q} \mid p \in \mathbb{Z}, q \in \mathbb{N}^+\right\} \\
\mathbb{R} &= \{x \mid x \text{ is a real number}\}
\end{aligned}
$$

Defining Sets Using Predicates

- Let $P(x)$ be a **predicate** specifying a property about x.
  For instance: "*x is an integer between 10 and 20*".
- The set of all x that satisfy the predicate $P(x)$ is written as

$$\{x \mid P(x)\} \quad \text{or} \quad \{x : P(x)\}$$

Defining Sets Using Predicates

- Let $P(x)$ be a **predicate** specifying a property about x.
  For instance: "*x is an integer between 10 and 20*".
- The set of all x that satisfy the predicate $P(x)$ is written as

$$\{x \mid P(x)\} \quad \text{or} \quad \{x : P(x)\}$$

- We may specify the **domain** from which x is taken, especially if ambiguity could arise. For example,

$$\{x \in \mathbb{Z} \mid 0 \leq x \leq 10\} \quad \text{vs} \quad \{x \in \mathbb{R} \mid 0 \leq x \leq 10\}$$

The Empty Set and Subsets

- The empty set (notation: $\emptyset$ or $\{\ \}$) is defined as the set that has no elements.

$$\forall x(x \notin \emptyset).$$

The Empty Set and Subsets

- The empty set (notation: $\emptyset$ or $\{\ \}$) is defined as the set that has no elements.

$$\forall x(x \notin \emptyset).$$

- Given sets S and T, S is a subset of T, or S is contained in T, (notation: $S \subseteq T$) iff each element of S is also an element of T.
- More precisely:

$$S \subseteq T \quad \text{iff} \quad \forall x\,(x \in S \rightarrow x \in T)$$

Above, $\forall$ denotes the **universal quantifier**, read "for all".

The Empty Set and Subsets

- The empty set (notation: $\emptyset$ or $\{\ \}$) is defined as the set that has no elements.

$$\forall x(x \notin \emptyset).$$

- Given sets S and T, S is a subset of T, or S is contained in T, (notation: $S \subseteq T$) iff each element of S is also an element of T.
- More precisely:

$$S \subseteq T \quad \text{iff} \quad \forall x\,(x \in S \rightarrow x \in T)$$

Above, $\forall$ denotes the **universal quantifier**, read "for all".

- For every set S, we have $S \subseteq S$ and $\emptyset \subseteq S$.

The Empty Set and Subsets

- The empty set (notation: $\emptyset$ or $\{\ \}$) is defined as the set that has no elements.

$$\forall x(x \notin \emptyset).$$

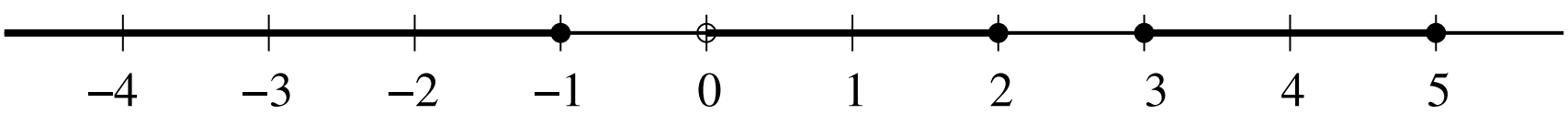
- Given sets S and T, S is a subset of T, or S is contained in T, (notation: $S \subseteq T$) iff each element of S is also an element of T.
- More precisely:

$$S \subseteq T \quad \text{iff} \quad \forall x\,(x \in S \rightarrow x \in T)$$

Above, $\forall$ denotes the **universal quantifier**, read "for all".

- For every set S, we have $S \subseteq S$ and $\emptyset \subseteq S$.
- Set S is a proper/strict subset of T ($S \subsetneq T$) if $S \subseteq T$ and $S \neq T$. (Avoid notation $S \subset T$, ambiguous)
- Examples: $\mathbb{Z}^+ \subsetneq \mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q} \subsetneq \mathbb{R} \subsetneq \mathbb{C}$.

Intervals of $\mathbb{R}$



A subset of $\mathbb{R}$ defined by two real numbers a and b, with the property that any real between a and b is included in the set.

- $[a, b] = \{x \in \mathbb{R} \mid a \leqslant x \leqslant b\}$
- $(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$
- $[a, b) = \{x \in \mathbb{R} \mid a \leqslant x < b\}$
- $(a, b] = \{x \in \mathbb{R} \mid a < x \leqslant b\}$
- $(-\infty, a] = \{x \in \mathbb{R} \mid x \leqslant a\}$
- $(a, a] = [a, a) = (a, a) = \emptyset$
- $a \geq b \Rightarrow [a, b) = (a, b] = (a, b) = \emptyset$
- ...

Equality of Sets

- Two sets A and B are equal if they contain the same elements. Formally,

$$A = B \quad \text{iff} \quad \forall x(x \in A \leftrightarrow x \in B).$$

Equality of Sets

- Two sets A and B are equal if they contain the same elements. Formally,

$$A = B \quad \text{iff} \quad \forall x(x \in A \leftrightarrow x \in B).$$

- We have:

$$A = B \quad \text{iff} \quad A \subseteq B \text{ and } B \subseteq A$$

Often used when proving that two sets are equal.

Sets of Sets

- Elements of a set may be sets themselves. For instance:

$$\{a, b, \{a\}, \{a, b\}\}$$

- Note: the set $\{a\}$ is not the same as $\{\{a\}\}$

Sets of Sets

- Elements of a set may be sets themselves. For instance:

$$\{a, b, \{a\}, \{a, b\}\}$$

- Note: the set $\{a\}$ is not the same as $\{\{a\}\}$
  Question: Do we have: $\{a\} \subseteq \{\{a\}\}$?
  Do we have: $\{a\} \in \{\{a\}\}$?

Sets of Sets

- Elements of a set may be sets themselves. For instance:

$$\{a, b, \{a\}, \{a, b\}\}$$

- Note: the set $\{a\}$ is not the same as $\{\{a\}\}$
  Question: Do we have: $\{a\} \subseteq \{\{a\}\}$? No, because $a \notin \{\{a\}\}$
  Do we have: $\{a\} \in \{\{a\}\}$?

Sets of Sets

- Elements of a set may be sets themselves. For instance:

$$\{a, b, \{a\}, \{a, b\}\}$$

- Note: the set $\{a\}$ is not the same as $\{\{a\}\}$
  Question: Do we have: $\{a\} \subseteq \{\{a\}\}$? No, because $a \notin \{\{a\}\}$
  Do we have: $\{a\} \in \{\{a\}\}$? Yes!

Sets of Sets

- Elements of a set may be sets themselves. For instance:

$$\{a, b, \{a\}, \{a, b\}\}$$

- Note: the set $\{a\}$ is not the same as $\{\{a\}\}$
  Question: Do we have: $\{a\} \subseteq \{\{a\}\}$? No, because $a \notin \{\{a\}\}$
  Do we have: $\{a\} \in \{\{a\}\}$? Yes!

- The power set $\mathcal{P}(S)$ of S is the set of all subsets of S:

$$\mathcal{P}(S) = \{T \mid T \subseteq S\}$$

Question: For $S = \{a\}$, what is $\mathcal{P}(S)$?

Sets of Sets

- Elements of a set may be sets themselves. For instance:

$$\{a, b, \{a\}, \{a, b\}\}$$

- Note: the set $\{a\}$ is not the same as $\{\{a\}\}$
  Question: Do we have: $\{a\} \subseteq \{\{a\}\}$? No, because $a \notin \{\{a\}\}$
  Do we have: $\{a\} \in \{\{a\}\}$? Yes!

- The power set $\mathcal{P}(S)$ of S is the set of all subsets of S:

$$\mathcal{P}(S) = \{T \mid T \subseteq S\}$$

Question: For $S = \{a\}$, what is $\mathcal{P}(S)$? $\mathcal{P}(\{a\}) = \{\emptyset, \{a\}\}$.

Size / Cardinality

- The cardinality (or size) of a set S is denoted as $|S|$.
- A **finite** set is a set that has a finite number of elements. If S is finite, then $|S|$ is the number of elements it contains.
- A set that is not finite is called **infinite**.
- Example: the set $S = \{a, b, c\}$ is finite and $|S| = 3$. The interval $[1, 3]$ is infinite. So are $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{R}$.

Size / Cardinality

- The cardinality (or size) of a set S is denoted as $|S|$.
- A **finite** set is a set that has a finite number of elements. If S is finite, then $|S|$ is the number of elements it contains.
- A set that is not finite is called **infinite**.
- Example: the set $S = \{a, b, c\}$ is finite and $|S| = 3$. The interval $[1, 3]$ is infinite. So are $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{R}$.
- Let S be a finite set. The cardinality of $\mathcal{P}(S)$ is $2^{|S|}$.
  For instance, if $|S| = 3$ then $|\mathcal{P}(S)| = 2^3 = 8$.

Size / Cardinality

- The cardinality (or size) of a set S is denoted as $|S|$.
- A **finite** set is a set that has a finite number of elements. If S is finite, then $|S|$ is the number of elements it contains.
- A set that is not finite is called **infinite**.
- Example: the set $S = \{a, b, c\}$ is finite and $|S| = 3$. The interval $[1, 3]$ is infinite. So are $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{R}$.
- Let S be a finite set. The cardinality of $\mathcal{P}(S)$ is $2^{|S|}$.
  For instance, if $|S| = 3$ then $|\mathcal{P}(S)| = 2^3 = 8$.
- Cardinalities of infinite sets: more later.

Set Universe

To avoid paradoxes (such as Russell's paradox)

- Only use relative abstraction $\{x \in A \mid P(x)\}$ (where A is a set) as a principle to construct sets.
- Or, we consider a context which is given by a well-defined universe U of objects of interest (which can be sets themselves). Any set within this context is a subset of U. For example, $U = \mathbb{R}$ if we are interested in collections of real numbers.



From "Logicomix".

Outline

1 Intro

2 Sets and Subsets

3 Operations on Sets

4 Sequences

5 String and Languages

6 Integers

7 Matrices

8 Mathematical Structures

Set Operations (1)

Let U denote the universe, and let A and B be sets in this universe.

Union: $A \cup B = \{x \mid x \in A \ \vee \ x \in B\}$

that is, for all $x \in U$: $x \in A \cup B$ iff $x \in A$ or $x \in B$

Set Operations (1)

Let U denote the universe, and let A and B be sets in this universe.

Union: $A \cup B = \{x \mid x \in A \ \vee \ x \in B\}$

that is, for all $x \in U$: $x \in A \cup B$ iff $x \in A$ or $x \in B$

Intersection: $A \cap B = \{x \mid x \in A \ \wedge \ x \in B \}$

that is, for all $x \in U$: $x \in A \cap B$ iff $x \in A$ and $x \in B$

Set Operations (1)

Let U denote the universe, and let A and B be sets in this universe.

Union: $A \cup B = \{x \mid x \in A \ \vee \ x \in B\}$

that is, for all $x \in U$: $x \in A \cup B$ iff $x \in A$ or $x \in B$

Intersection: $A \cap B = \{x \mid x \in A \ \wedge \ x \in B \}$

that is, for all $x \in U$: $x \in A \cap B$ iff $x \in A$ and $x \in B$

Notation:

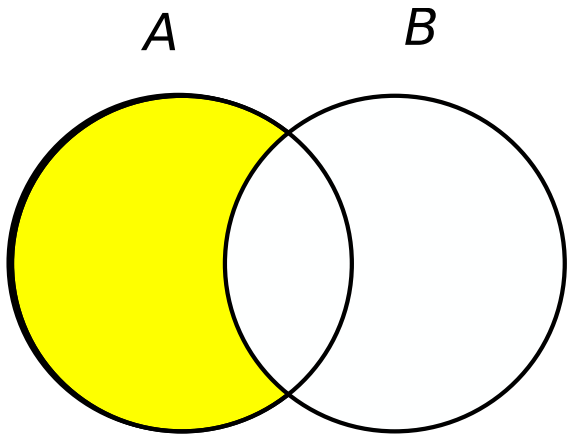
$$\bigcup_{i=1}^{n} A_i = A_1 \cup \ldots \cup A_n$$

Notation:

$$\bigcap_{i=1}^{n} A_i = A_1 \cap \ldots \cap A_n$$

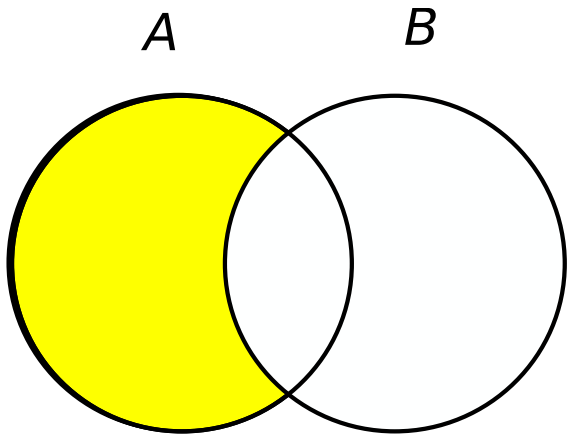
Set Operations (2)

- **Relative complement of B in A** or **set difference**:
  $A - B = \{x \mid x \in A \ \wedge \ x \notin B\}$. Alternative notation: $A \setminus B$.
  Informally, $A - B$ is the set of all x that are in A, but not in B.



Set Operations (2)

- **Relative complement of B in A** or **set difference**:
  $A - B = \{x \mid x \in A \ \wedge \ x \notin B\}$. Alternative notation: $A \setminus B$.
  Informally, $A - B$ is the set of all x that are in A, but not in B.



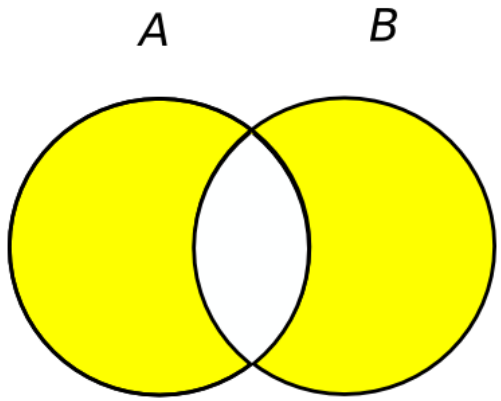
- **Complement**: $\overline{A} = U - A = U \setminus A$. Alternative notation: A^c.
  Informally, $\overline{A}$ is the set of all x that are not in A.

Set Operations (3)

- **Symmetric difference**:

$$\begin{aligned} A \oplus B &= \{x \mid (x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A)\} \\ &= (A - B) \cup (B - A) \end{aligned}$$

Alternative notation: $A \Delta B$.



Algebraic Laws of Set Operations (1)

Commutativity		$A \cup B = B \cup A$
		$A \cap B = B \cap A$
Associativity		$(A \cup B) \cup C = A \cup (B \cup C)$
		$(A \cap B) \cap C = A \cap (B \cap C)$
Distributivity	($\cup$ over $\cap$)	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
	($\cap$ over $\cup$)	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Algebraic Laws of Set Operations (2)

Idempotence	$A \cup A = A$ $A \cap A = A$
Empty set **Universal set**	$A \cup \emptyset = A, \quad A \cap \emptyset = \emptyset$ $A \cap U = A, \quad A \cup U = U$
Complement	$A \cup \overline{A} = U$ $A \cap \overline{A} = \emptyset$
Double Complement	$\overline{(\overline{A})} = A$
Relation universe - Empty set	$\overline{U} = \emptyset$ $\overline{\emptyset} = U$
De Morgan Laws	$\overline{(A \cup B)} = \overline{A} \cap \overline{B}$ $\overline{(A \cap B)} = \overline{A} \cup \overline{B}$

Annotated Linear Proof (ALP)

(Also called: chain-of-equivalences proof.)

Proposition: $A \cup (B \cup C) = (A \cup B) \cup C$

Proof By the definition of set equality, we must show that:

$$\text{for all } x: \quad x \in A \cup (B \cup C) \quad \text{iff} \quad x \in (A \cup B) \cup C.$$

[1] Kolman et al. write $\equiv$ instead of $\Leftrightarrow$. Alternatively, you can write "iff".

Annotated Linear Proof (ALP)

(Also called: chain-of-equivalences proof.)

Proposition: $A \cup (B \cup C) = (A \cup B) \cup C$

Proof By the definition of set equality, we must show that:

$$\text{for all } x: \quad x \in A \cup (B \cup C) \quad \text{iff} \quad x \in (A \cup B) \cup C.$$

So let x be arbitrary. We have[1]:

$x \in A \cup (B \cup C)$ (start with LHS), apply reasoning...

[1] Kolman et al. write $\equiv$ instead of $\Leftrightarrow$. Alternatively, you can write "iff".

Annotated Linear Proof (ALP)

(Also called: chain-of-equivalences proof.)

Proposition: $A \cup (B \cup C) = (A \cup B) \cup C$

Proof By the definition of set equality, we must show that:

$$\text{for all } x: \quad x \in A \cup (B \cup C) \quad \text{iff} \quad x \in (A \cup B) \cup C.$$

So let x be arbitrary. We have[1]:

	$x \in A \cup (B \cup C)$	(start with LHS), apply reasoning...
$\Leftrightarrow$	$(x \in A) \vee (x \in B \cup C)$	(definition $\cup$)

[1] Kolman et al. write $\equiv$ instead of $\Leftrightarrow$. Alternatively, you can write "iff".

Annotated Linear Proof (ALP)

(Also called: chain-of-equivalences proof.)

Proposition: $A \cup (B \cup C) = (A \cup B) \cup C$

Proof By the definition of set equality, we must show that:

$$\text{for all } x: \quad x \in A \cup (B \cup C) \quad \text{iff} \quad x \in (A \cup B) \cup C.$$

So let x be arbitrary. We have[1]:

	$x \in A \cup (B \cup C)$	(start with LHS), apply reasoning...
$\Leftrightarrow$	$(x \in A) \vee (x \in B \cup C)$	(definition $\cup$)
$\Leftrightarrow$	$x \in A \vee (x \in B \vee x \in C)$	(definition $\cup$)

[1] Kolman et al. write $\equiv$ instead of $\Leftrightarrow$. Alternatively, you can write "iff".

Annotated Linear Proof (ALP)

(Also called: chain-of-equivalences proof.)

Proposition: $A \cup (B \cup C) = (A \cup B) \cup C$

Proof By the definition of set equality, we must show that:

$$\text{for all } x: \quad x \in A \cup (B \cup C) \quad \text{iff} \quad x \in (A \cup B) \cup C.$$

So let x be arbitrary. We have[1]:

	$x \in A \cup (B \cup C)$	(start with LHS), apply reasoning...
$\Leftrightarrow$	$(x \in A) \vee (x \in B \cup C)$	(definition $\cup$)
$\Leftrightarrow$	$x \in A \vee (x \in B \vee x \in C)$	(definition $\cup$)
$\Leftrightarrow$	$(x \in A \vee x \in B) \vee x \in C$	(logic, associativity of $\vee$)

[1] Kolman et al. write $\equiv$ instead of $\Leftrightarrow$. Alternatively, you can write "iff".

Annotated Linear Proof (ALP)

(Also called: chain-of-equivalences proof.)

Proposition: $A \cup (B \cup C) = (A \cup B) \cup C$

Proof By the definition of set equality, we must show that:

$$\text{for all } x: \quad x \in A \cup (B \cup C) \quad \text{iff} \quad x \in (A \cup B) \cup C.$$

So let x be arbitrary. We have[1]:

	$x \in A \cup (B \cup C)$	(start with LHS), apply reasoning...
$\Leftrightarrow$	$(x \in A) \vee (x \in B \cup C)$	(definition $\cup$)
$\Leftrightarrow$	$x \in A \vee (x \in B \vee x \in C)$	(definition $\cup$)
$\Leftrightarrow$	$(x \in A \vee x \in B) \vee x \in C$	(logic, associativity of $\vee$)
$\Leftrightarrow$	$(x \in A \cup B) \vee (x \in C)$	(definition $\cup$)

[1] Kolman et al. write $\equiv$ instead of $\Leftrightarrow$. Alternatively, you can write "iff".

Annotated Linear Proof (ALP)

(Also called: chain-of-equivalences proof.)

Proposition: $A \cup (B \cup C) = (A \cup B) \cup C$

Proof By the definition of set equality, we must show that:

$$\text{for all } x: \quad x \in A \cup (B \cup C) \quad \text{iff} \quad x \in (A \cup B) \cup C.$$

So let x be arbitrary. We have[1]:

	$x \in A \cup (B \cup C)$	(start with LHS), apply reasoning...
$\Leftrightarrow$	$(x \in A) \vee (x \in B \cup C)$	(definition $\cup$)
$\Leftrightarrow$	$x \in A \vee (x \in B \vee x \in C)$	(definition $\cup$)
$\Leftrightarrow$	$(x \in A \vee x \in B) \vee x \in C$	(logic, associativity of $\vee$)
$\Leftrightarrow$	$(x \in A \cup B) \vee (x \in C)$	(definition $\cup$)
$\Leftrightarrow$	$x \in (A \cup B) \cup C$	(definition $\cup$)
	(...until we reach RHS)	

[1] Kolman et al. write $\equiv$ instead of $\Leftrightarrow$. Alternatively, you can write "iff".

Outline

1 Intro

2 Sets and Subsets

3 Operations on Sets

4 Sequences

5 String and Languages

6 Integers

7 Matrices

8 Mathematical Structures

Sequences

- A sequence is a **list** of objects in a definite order.
- Example: $1, 2, 3, 5, 7$ (a finite sequence of length 5)
- Example: m,a,t,h,e,m,a,t,i,c,s (a finite sequence of length 11)
- Example: $0, 2, 4, 6, 8, 10, 12, \ldots$ (an infinite sequence)
- Note: elements in a sequence can occur multiple times, and order matters. $1, 2, 3$ is not the same sequence as $2, 3, 1$.

[2] In CS, we usually start counting at 0, cf. Dijkstra: `https://www.cs.utexas.edu/users/EWD/transcriptions/EWD08xx/EWD831.html`.

Sequences

- A sequence is a **list** of objects in a definite order.
- Example: $1, 2, 3, 5, 7$ (a finite sequence of length 5)
- Example: m,a,t,h,e,m,a,t,i,c,s (a finite sequence of length 11)
- Example: $0, 2, 4, 6, 8, 10, 12, \ldots$ (an infinite sequence)
- Note: elements in a sequence can occur multiple times, and order matters. $1, 2, 3$ is not the same sequence as $2, 3, 1$.
- More formally: a sequence is an indexed list[2] (possibly infinite) $s = s_0, s_1, s_2, s_3, s_4, \ldots$ (or $s = s_1, s_2, s_3, s_4, s_5, \ldots$)
- Ex: $s = a, b, c, d, e, \ldots$ is a sequence, with $s_0 = a$, $s_1 = b$, etc.
- Note: Lists and arrays are datatypes. Sequences are mathematical objects.

[2] In CS, we usually start counting at 0, cf. Dijkstra: `https://www.cs.utexas.edu/users/EWD/transcriptions/EWD08xx/EWD831.html`.

Specifying Sequences

The notation $s = 1, 2, 3, 4, 5, \ldots$ is suggestive, but imprecise.

Specifying Sequences

The notation $s = 1, 2, 3, 4, 5, \ldots$ is suggestive, but imprecise.

Recursive definition

- Specify a starting value (s_0)
- Specify s_n using previous values.
- Example: $s_0 = 3$ and $s_n = 2 \cdot s_{n-1} + 7$ for all $n \in \mathbb{N}, n \geq 1$ defines the infinite sequence $3, 13, 33, 73, \ldots$

Specifying Sequences

The notation $s = 1, 2, 3, 4, 5, \ldots$ is suggestive, but imprecise.

Recursive definition

- Specify a starting value (s_0)
- Specify s_n using previous values.
- Example: $s_0 = 3$ and $s_n = 2 \cdot s_{n-1} + 7$ for all $n \in \mathbb{N}, n \geq 1$ defines the infinite sequence $3, 13, 33, 73, \ldots$

Explicit definition

- Specify each s_n as a function of n.
- Example: $s_n = n^2$, $n \geq 0$, specifies the infinite sequence $0, 1, 4, 9, 16, 25, \ldots$.

Characteristic Function

- Let A be a subset of the universe U.
  The **characteristic function** (or indicator function) of A
  $f_A : U \to \{0, 1\}$ is defined (for all $x \in U$) as:

$$f_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

- Alternative common notation: $\chi_A(x)$ (the Greek letter χ "chi"))

Computer Representation of Sets and Subsets

- Each finite universe U can be represented by a finite sequence by choosing an ordering of its elements.
- For example, the set $U = \{c, a, s, q, t\}$ can be represented by the sequence $\alpha = a, c, q, s, t$.

Computer Representation of Sets and Subsets

- Each finite universe U can be represented by a finite sequence by choosing an ordering of its elements.
- For example, the set $U = \{c, a, s, q, t\}$ can be represented by the sequence $\alpha = a, c, q, s, t$.
- Now, we can represent a subset $A \subseteq U$ as an array with entries 0 and 1 using the characteristic function of A.
- Examples:

	a	c	q	s	t
Representation of U:	1	1	1	1	1
Representation of $A = \{c, s\}$:	0	1	0	1	0

Computer Representation of Sets and Subsets

- Each finite universe U can be represented by a finite sequence by choosing an ordering of its elements.
- For example, the set $U = \{c, a, s, q, t\}$ can be represented by the sequence $\alpha = a, c, q, s, t$.
- Now, we can represent a subset $A \subseteq U$ as an array with entries 0 and 1 using the characteristic function of A.
- Examples:

	a	c	q	s	t
Representation of U:	1	1	1	1	1
Representation of $A = \{c, s\}$:	0	1	0	1	0

- More precisely, a subset $A \subseteq U$ is represented by an array $\mathtt{A}$ of length $|U|$ where $\mathtt{A}[k] = f_A(\alpha_k)$ for $k = 0, \ldots, |U| - 1$.

Countable and Uncountable Sets

- A set S is called **countable** (or denumerable) if all elements of S can be written as a sequence.
- Each finite set is countable.
- Examples: sets $\mathbb{N}$, $\mathbb{Z}$ and $\mathbb{Q}$ are countable.
- Infinite sets that are not countable are called **uncountable**.

[3] A proof method devised in 1891 by Georg Cantor. Similar ideas used by Gödel and Turing.

Countable and Uncountable Sets

- A set S is called **countable** (or denumerable) if all elements of S can be written as a sequence.
- Each finite set is countable.
- Examples: sets $\mathbb{N}$, $\mathbb{Z}$ and $\mathbb{Q}$ are countable.
- Infinite sets that are not countable are called **uncountable**.
- The set $\mathbb{R}$ is not countable. The proof (see the book) uses Cantor's diagonal-argument.[3]
- The set $\mathcal{P}(\mathbb{N})$ is also uncountable (in fact, it has the same cardinality as $\mathbb{R}$).

[3] A proof method devised in 1891 by Georg Cantor. Similar ideas used by Gödel and Turing.

Outline

1 Intro

2 Sets and Subsets

3 Operations on Sets

4 Sequences

5 String and Languages

6 Integers

7 Matrices

8 Mathematical Structures

Formal Languages

- **Alphabet**: non-empty finite set Σ of **letters** (or symbols).
- **Word/String over** Σ: a finite sequence of letters from alphabet Σ.
- Instead of $a_1, a_2, a_3, \ldots, a_n$, we usually write it as $a_1 a_2 a_3 \ldots a_n$, and say $a_1 a_2 a_3 \ldots a_n$ is a word of **length** n.
- Example: $b, l, a, b, l, a, b, l, a$ is written *blablabla* (length 9).

[4] Kolman et al. write Λ for the empty string, which is nonstandard!

Formal Languages

- **Alphabet**: non-empty finite set Σ of **letters** (or symbols).
- **Word/String over** Σ: a finite sequence of letters from alphabet Σ.
- Instead of $a_1, a_2, a_3, \ldots, a_n$, we usually write it as $a_1 a_2 a_3 \ldots a_n$, and say $a_1 a_2 a_3 \ldots a_n$ is a word of **length** n.
- Example: $b, l, a, b, l, a, b, l, a$ is written *blablabla* (length 9).
- The **empty word/string**, usually[4] written $\epsilon \in \Sigma^*$, has length 0.

[4] Kolman et al. write Λ for the empty string, which is nonstandard!

Formal Languages

- **Alphabet**: non-empty finite set Σ of **letters** (or symbols).
- **Word/String over** Σ: a finite sequence of letters from alphabet Σ.
- Instead of $a_1, a_2, a_3, \ldots, a_n$, we usually write it as $a_1 a_2 a_3 \ldots a_n$, and say $a_1 a_2 a_3 \ldots a_n$ is a word of **length** n.
- Example: $b, l, a, b, l, a, b, l, a$ is written *blablabla* (length 9).
- The **empty word/string**, usually[4] written $\epsilon \in \Sigma^*$, has length 0.
- Σ^* denotes the set of all words over Σ

[4] Kolman et al. write Λ for the empty string, which is nonstandard!

Formal Languages

- **Alphabet**: non-empty finite set Σ of **letters** (or symbols).
- **Word/String over** Σ: a finite sequence of letters from alphabet Σ.
- Instead of $a_1, a_2, a_3, \ldots, a_n$, we usually write it as $a_1 a_2 a_3 \ldots a_n$, and say $a_1 a_2 a_3 \ldots a_n$ is a word of **length** n.
- Example: $b, l, a, b, l, a, b, l, a$ is written *blablabla* (length 9).
- The **empty word/string**, usually[4] written $\epsilon \in \Sigma^*$, has length 0.
- Σ^* denotes the set of all words over Σ
- **(Formal) Language:** a subset $L \subseteq \Sigma^*$ (i.e., L is a set of words)

[4] Kolman et al. write Λ for the empty string, which is nonstandard!

Formal Languages

- **Alphabet**: non-empty finite set Σ of **letters** (or symbols).
- **Word/String over** Σ: a finite sequence of letters from alphabet Σ.
- Instead of $a_1, a_2, a_3, \ldots, a_n$, we usually write it as $a_1 a_2 a_3 \ldots a_n$, and say $a_1 a_2 a_3 \ldots a_n$ is a word of **length** n.
- Example: $b, l, a, b, l, a, b, l, a$ is written *blablabla* (length 9).
- The **empty word/string**, usually[4] written $\epsilon \in \Sigma^*$, has length 0.
- Σ^* denotes the set of all words over Σ
- **(Formal) Language:** a subset $L \subseteq \Sigma^*$ (i.e., L is a set of words)

Example

- Let $\Sigma = \{a\}$.
- Some words in Σ^*: a, aa, aaa.
- The set of all words over Σ is $\Sigma^* = \{\epsilon, a, aa, aaa, aaaa, \ldots\}$

[4] Kolman et al. write Λ for the empty string, which is nonstandard!

Concatenation of Words

- Consider the two words

$$w_1 = \text{itisrainingcats}$$
$$w_2 = \text{anddogs}$$

- Their **concatenation** is

$$w_1 \cdot w_2 = \text{itisrainingcatsanddogs}$$

- We often write $w_1 w_2$ instead of $w_1 \cdot w_2$
- Note: $w\epsilon = \epsilon w = w$ for all words w.
- Concatenation is **associative**:

$$(w_1 w_2) w_3 = w_1 (w_2 w_3)$$

So we usually omit parentheses: $w_1 w_2 w_3$

Language Concatenation

- Consider the two languages over $\Sigma = \{a, b, c\}$:

$$L = \{aa, bb\}, \quad M = \{\epsilon, c, ba\}$$

- Their **concatenation** is

$$L \cdot M = \{aa, bb, aac, bbc, aaba, bbba\}$$

- More generally, the concatenation of two arbitrary languages $L, M \in \mathcal{P}(\Sigma^*)$ is the language:

$$L \cdot M = LM = \{s \cdot t \mid s \in L \wedge t \in M\}$$

Kleene Star of Languages

- For a language $L \subseteq \Sigma^*$, L^* is the set of all words that can be built by concatenating zero or more words from L.
- That is, think of L as an alphabet over which we can build new words.
- Example:

$$\begin{aligned} L &= \{aa, b\} \\ L^* &= \{\epsilon, aa, b, aaaa, aab, baa, bb, \ldots\} \end{aligned}$$

Regular Expressions

Let Σ be an alphabet. A **regular expression** (regexp) over Σ is defined recursively/inductively in terms of members of the alphabet, and the symbols $($ $)$ $\cdot$ $\vee$ * ϵ according to the following rules:

Regular Expressions

Let Σ be an alphabet. A **regular expression** (regexp) over Σ is defined recursively/inductively in terms of members of the alphabet, and the symbols $($ $)$ $\cdot$ $\vee$ * ϵ according to the following rules:

- The symbol ϵ is a regular expression.
- Each $a \in \Sigma$ is a regular expression.

Regular Expressions

Let Σ be an alphabet. A **regular expression** (regexp) over Σ is defined recursively/inductively in terms of members of the alphabet, and the symbols $($ $)$ $\cdot$ $\vee$ * ϵ according to the following rules:

- The symbol ϵ is a regular expression.
- Each $a \in \Sigma$ is a regular expression.
- If α and β are regular expressions, then $(\alpha \cdot \beta)$ and $(\alpha \vee \beta)$ are regular expressions
- If α is a regular expression, then $(\alpha)^*$ is a regular expression.

Regular Expressions

Let Σ be an alphabet. A **regular expression** (regexp) over Σ is defined recursively/inductively in terms of members of the alphabet, and the symbols $($ $)$ $\cdot$ $\vee$ * ϵ according to the following rules:

- The symbol ϵ is a regular expression.
- Each $a \in \Sigma$ is a regular expression.
- If α and β are regular expressions, then $(\alpha \cdot \beta)$ and $(\alpha \vee \beta)$ are regular expressions
- If α is a regular expression, then $(\alpha)^*$ is a regular expression.
- Nothing else is a regular expression.

Regular Expressions

Let Σ be an alphabet. A **regular expression** (regexp) over Σ is defined recursively/inductively in terms of members of the alphabet, and the symbols $($ $)$ $\cdot$ $\vee$ * ϵ according to the following rules:

- The symbol ϵ is a regular expression.
- Each $a \in \Sigma$ is a regular expression.
- If α and β are regular expressions, then $(\alpha \cdot \beta)$ and $(\alpha \vee \beta)$ are regular expressions
- If α is a regular expression, then $(\alpha)^*$ is a regular expression.
- Nothing else is a regular expression.

Examples: $\Sigma = \{a, b\}$.
Some regexps over Σ: $abab$, $(a^* \vee aba)^*$, $(a \vee \epsilon)^*(a^* b)$

Regular Sets (set-theoretic interpretation of regexps)

Each regexp over Σ corresponds to a subset of Σ^*, i.e., a language.

- ϵ corresponds to the set $\{\epsilon\}$.
- $a \in \Sigma$ corresponds to the set $\{a\}$.
- If α and β are regexps that correspond to sets $M, N \subseteq \Sigma^*$, then:
  - $(\alpha \cdot \beta)$ corresponds to the set $M \cdot N = \{s \cdot t \mid s \in M \wedge t \in N\}$
  - $(\alpha \vee \beta)$ corresponds to the set $M \cup N$.
  - $(\alpha)^*$ corresponds to set M^*, i.e., all words/strings over M.

Regular Sets (set-theoretic interpretation of regexps)

Each regexp over Σ corresponds to a subset of Σ^*, i.e., a language.

- ϵ corresponds to the set $\{\epsilon\}$.
- $a \in \Sigma$ corresponds to the set $\{a\}$.
- If α and β are regexps that correspond to sets $M, N \subseteq \Sigma^*$, then:
  - $(\alpha \cdot \beta)$ corresponds to the set $M \cdot N = \{s \cdot t \mid s \in M \wedge t \in N\}$
  - $(\alpha \vee \beta)$ corresponds to the set $M \cup N$.
  - $(\alpha)^*$ corresponds to set M^*, i.e., all words/strings over M.

Exercise Let L be the set corresponding to $(a \vee b)^*(c \vee \epsilon)$.

True or false:

1. $abbac \in L$?
2. $abccc \in L$?
3. $abaa \in L$?
4. $c \in L$?
5. $\epsilon \in L$?

Regular Sets (set-theoretic interpretation of regexps)

Each regexp over Σ corresponds to a subset of Σ^*, i.e., a language.

- ϵ corresponds to the set $\{\epsilon\}$.
- $a \in \Sigma$ corresponds to the set $\{a\}$.
- If α and β are regexps that correspond to sets $M, N \subseteq \Sigma^*$, then:
  - $(\alpha \cdot \beta)$ corresponds to the set $M \cdot N = \{s \cdot t \mid s \in M \wedge t \in N\}$
  - $(\alpha \vee \beta)$ corresponds to the set $M \cup N$.
  - $(\alpha)^*$ corresponds to set M^*, i.e., all words/strings over M.

Exercise Let L be the set corresponding to $(a \vee b)^*(c \vee \epsilon)$.

True or false:

1. $abbac \in L$?
2. $abccc \in L$?
3. $abaa \in L$?
4. $c \in L$?
5. $\epsilon \in L$?

More in Languages and Machines (2nd yr). Applic: text search.

Outline

1 Intro

2 Sets and Subsets

3 Operations on Sets

4 Sequences

5 String and Languages

6 Integers

7 Matrices

8 Mathematical Structures

Euclidean Division of Integers

Theorem (Euclidean division lemma)

Let m, n be integers, with $n > 0$.
There are unique integers q and r, such that:

$$m = q \cdot n + r \quad \text{and} \quad 0 \leqslant r < n$$

where q is called **quotient**, r is called **remainder**.

Euclidean Division of Integers

Theorem (Euclidean division lemma)

Let m, n be integers, with $n > 0$.
There are unique integers q and r, such that:

$$m = q \cdot n + r \quad \text{and} \quad 0 \leqslant r < n$$

where q is called **quotient**, r is called **remainder**.

We can find them using the operators **div** and **mod**:

- $q = m$ **div** n (quotient of m with n)
- $r = m$ **mod** n (m modulo n)

Examples:			
	quotient	7 **div** $2 = 3$	-13 **div** $5 = -3$
	remainder	7 **mod** $2 = 1$	-13 **mod** $5 = 2$
		$7 = 3 \cdot 2 + 1$	$-13 = -3 \cdot 5 + 2$

Factors and Products (1)

We consider the set of integers $\mathbb{Z}$.

- We write $a \mid b$ if $b = k \cdot a$, for some integer k.
  We say that a is a **divisor** of b, or that b is a **multiple** of a.
- Example: $7 \mid 98$, since $98 = 7 \cdot 14$.

Factors and Products (1)

We consider the set of integers $\mathbb{Z}$.

- We write $a \mid b$ if $b = k \cdot a$, for some integer k.
  We say that a is a **divisor** of b, or that b is a **multiple** of a.
- Example: $7 \mid 98$, since $98 = 7 \cdot 14$.

Some properties (Theorem 2):

- If $(a \mid b)$ and $(a \mid c)$ then $a \mid (b + c)$
- If $(a \mid b)$ and $(a \mid c)$ where $b > c$ then $a \mid (b - c)$
- If $(a \mid b)$ or $(a \mid c)$ then $a \mid bc$
- If $(a \mid b)$ and $(b \mid c)$ then $a \mid c$ (“ $\mid$ ” is **transitive**)

Factors and Products (2)

Definition

*A positive natural number n is a **prime number** if it has exactly **two** unique divisors: 1 and n.*

Thus, by definition, 1 is not a prime. Still, 2, 3, 5, and 7 are prime.

Factors and Products (2)

Definition

*A positive natural number n is a **prime number** if it has exactly **two** unique divisors: 1 and n.*

Thus, by definition, 1 is not a prime. Still, 2, 3, 5, and 7 are prime.

Theorem (Euclid): There are infinitely many primes.

Factors and Products (2)

Definition

*A positive natural number n is a **prime number** if it has exactly **two** unique divisors: 1 and n.*

Thus, by definition, 1 is not a prime. Still, 2, 3, 5, and 7 are prime.

Theorem (Euclid): There are infinitely many primes.

Proof:

Factors and Products (2)

Definition

*A positive natural number n is a **prime number** if it has exactly **two** unique divisors: 1 and n.*

Thus, by definition, 1 is not a prime. Still, 2, 3, 5, and 7 are prime.

Theorem (Euclid): There are infinitely many primes.

Proof: Show that given any finite list of primes $p_1, \ldots, p_n$, there is always a prime not in the list. Let $n := p_1 \cdot \ldots \cdot p_n$, and consider $q := n + 1$. *Case q is prime:* Then q is not in the list, because $q > p_i$ for all p_i in the list. *Case q is not prime:* Then there is a prime p such that $1 < p < q$ and $p|q$. If p is in the list, then $p|n$, hence $p|(q - n) = 1$ which implies that $p = 1$, contradiction, since p is prime. Hence p cannot be in the list. In both cases, there is a prime not in the list.

See also `https://primes.utm.edu/notes/proofs/infinite/index.html`

Factors and Products (3)

Theorem (Prime factorization)

Any positive integer $n > 1$ can be uniquely written as a product of distinct prime numbers $p_1^{k_1} \cdot p_2^{k_2} \cdots p_s^{k_s}$, where $k_1, \ldots, k_s$ are positive integers.

Examples:

$$\begin{aligned} 60 &= 2 \cdot 2 \cdot 3 \cdot 5 &&= 2^2 \cdot 3 \cdot 5 \\ 168 &= 2 \cdot 2 \cdot 2 \cdot 3 \cdot 7 &&= 2^3 \cdot 3 \cdot 7 \\ 192 &= 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 3 &&= 2^6 \cdot 3 \end{aligned}$$

Factors and Products (3)

Theorem (Prime factorization)

Any positive integer $n > 1$ can be uniquely written as a product of distinct prime numbers $p_1^{k_1} \cdot p_2^{k_2} \cdots p_s^{k_s}$, where $k_1, \ldots, k_s$ are positive integers.

Examples:

$$\begin{aligned} 60 &= 2 \cdot 2 \cdot 3 \cdot 5 &&= 2^2 \cdot 3 \cdot 5 \\ 168 &= 2 \cdot 2 \cdot 2 \cdot 3 \cdot 7 &&= 2^3 \cdot 3 \cdot 7 \\ 192 &= 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 3 &&= 2^6 \cdot 3 \end{aligned}$$

Applications:

- Verifying prime factorization is easy: Prime factorization in NP.
- Finding prime factorization is hard (as far as we know). Basis for RSA public key cryptography. (But we don't know if NP-complete.)

GCD

Let a, b and k be positive integers.

- If ka and $k|b$, then k is a **common divisor** of a and b.
  The **greatest common divisor (GCD)** of a and b, written **gcd**(a, b), is the largest such k.
- Example: 3 and 2 are common divisors of 18 and 24.
  But **gcd**$(18, 24) = 6$.

GCD

Let a, b and k be positive integers.

- If ka and $k|b$, then k is a **common divisor** of a and b.
  The **greatest common divisor (GCD)** of a and b, written $\mathbf{gcd}(a, b)$, is the largest such k.
- Example: 3 and 2 are common divisors of 18 and 24.
  But $\mathbf{gcd}(18, 24) = 6$.
- If $\mathbf{gcd}(m, n) = 1$, then m and n are called **coprime** (or **relatively prime**). Example: 6 and 35 are coprime.

Bézout's Identity

For all $a, b \in \mathbb{Z}^+$, we have:

- $\gcd(a, b) = sa + tb$ for some integers $s, t \in \mathbb{Z}$.
- If $c|a$ and $c|b$, then $c|\gcd(a, b)$.

Euclid's Algorithm

Input: two positive integers a, b. **Output:** $\gcd(a, b)$.

Euclid's Algorithm

Input: two positive integers a, b. **Output:** $\gcd(a, b)$.
Determine quotient and remainder in successive steps, applying Euclidean Division Lemma. Assume $a > b$ (otherwise swap them).

$$\text{divide } a \text{ by } b \qquad a = q_1 b + r_1 \qquad 0 \leq r_1 < b$$

Euclid's Algorithm

Input: two positive integers a, b. **Output:** $\gcd(a, b)$.
Determine quotient and remainder in successive steps, applying Euclidean Division Lemma. Assume $a > b$ (otherwise swap them).

divide a by b	a	$=$	$q_1 b + r_1$	$0 \leq r_1 < b$
divide b by r_1	b	$=$	$q_2 r_1 + r_2$	$0 \leq r_2 < r_1$

Euclid's Algorithm

Input: two positive integers a, b. **Output:** $\gcd(a, b)$.
Determine quotient and remainder in successive steps, applying Euclidean Division Lemma. Assume $a > b$ (otherwise swap them).

divide a by b	a	$=$	$q_1 b + r_1$	$0 \leq r_1 < b$
divide b by r_1	b	$=$	$q_2 r_1 + r_2$	$0 \leq r_2 < r_1$
divide r_1 by r_2	r_1	$=$	$q_3 r_2 + r_3$	$0 \leq r_3 < r_2$

Euclid's Algorithm

Input: two positive integers a, b. **Output:** $\gcd(a, b)$.
Determine quotient and remainder in successive steps, applying Euclidean Division Lemma. Assume $a > b$ (otherwise swap them).

divide a by b	a	$=$	$q_1 b + r_1$	$0 \leq r_1 < b$
divide b by r_1	b	$=$	$q_2 r_1 + r_2$	$0 \leq r_2 < r_1$
divide r_1 by r_2	r_1	$=$	$q_3 r_2 + r_3$	$0 \leq r_3 < r_2$
$\ldots$			$\ldots$	$\ldots$
divide r_{n-1} by r_n	r_{n-1}	$=$	$q_{n+1} r_n + r_{n+1}$	$0 \leq r_{n+1} < r_n$

At some point $r_{n+1} = 0$. Claim: $\gcd(a, b) = r_n$.

Euclid's Algorithm

Input: two positive integers a, b. **Output:** $\gcd(a, b)$.
Determine quotient and remainder in successive steps, applying Euclidean Division Lemma. Assume $a > b$ (otherwise swap them).

$$\begin{array}{llcll}
\text{divide } a \text{ by } b & a & = & q_1 b + r_1 & 0 \leq r_1 < b \\
\text{divide } b \text{ by } r_1 & b & = & q_2 r_1 + r_2 & 0 \leq r_2 < r_1 \\
\text{divide } r_1 \text{ by } r_2 & r_1 & = & q_3 r_2 + r_3 & 0 \leq r_3 < r_2 \\
\ldots & & & \ldots & \ldots \\
\text{divide } r_{n-1} \text{ by } r_n & r_{n-1} & = & q_{n+1} r_n + r_{n+1} & 0 \leq r_{n+1} < r_n
\end{array}$$

At some point $r_{n+1} = 0$. Claim: $\gcd(a, b) = r_n$. Proof: By Theorem 2, if $k|a$ and $k|b$ then $k|a - q_1 b = r_1$. Repeat this argument to get:

$$\mathbf{gcd}(a, b) = \mathbf{gcd}(b, r_1) = \mathbf{gcd}(r_1, r_2) = \cdots = \mathbf{gcd}(r_{n-1}, r_n) = r_n$$

Example: $a = 24, b = 18$, $\mathbf{gcd}(24, 18) = \mathbf{gcd}(18, 6) = 6$

LCM

If $a \mid k$ and $b \mid k$, then k is a **common multiple** of a and b.

The smallest such k is the **least common multiple** (or LCM).
We write $\mathbf{lcm}(a, b) = k$. Example: $\mathbf{lcm}(18, 24) = 72$.

For any two positive natural numbers m and n we have:

- any common multiple of m and n is a multiple of $\mathbf{lcm}(m, n)$
- $\mathbf{gcd}(m, n) \cdot \mathbf{lcm}(m, n) = m \cdot n$

GCD and LCM (Theorem 6, Example)

Prime factorisations:
$$168 = 2^3 \cdot 3 \cdot 7 = 2^3 \cdot 3^1 \cdot 7^1$$
$$192 = 2^6 \cdot 3 = 2^6 \cdot 3^1 \cdot 7^0$$

- **Greatest common divisor (GCD)**:

$$\mathbf{gcd}(168, 192) = \mathbf{gcd}(2^3 \cdot 3 \cdot 7\,,\, 2^6 \cdot 3)$$
$$= 2^{min(3,6)} \cdot 3^{min(1,1)} \cdot 7^{min(1,0)} = 2^3 \cdot 3^1 \cdot 7^0 = 24$$

GCD and LCM (Theorem 6, Example)

Prime factorisations: $168 = 2^3 \cdot 3 \cdot 7 = 2^3 \cdot 3^1 \cdot 7^1$

$192 = 2^6 \cdot 3 = 2^6 \cdot 3^1 \cdot 7^0$

- **Greatest common divisor (GCD)**:

$$\begin{aligned}\textbf{gcd}(168, 192) &= \textbf{gcd}(2^3 \cdot 3 \cdot 7\,,\, 2^6 \cdot 3) \\ &= 2^{min(3,6)} \cdot 3^{min(1,1)} \cdot 7^{min(1,0)} = 2^3 \cdot 3^1 \cdot 7^0 = 24\end{aligned}$$

- **Least common multiple (LCM)**:

$$\begin{aligned}\textbf{lcm}(168, 192) &= \textbf{lcm}(2^3 \cdot 3 \cdot 7\,,\, 2^6 \cdot 3) \\ &= 2^{max(3,6)} \cdot 3^{max(1,1)} \cdot 7^{max(1,0)} = 2^6 \cdot 3^1 \cdot 7^1 = 1344\end{aligned}$$

GCD and LCM (Theorem 6, Example)

Prime factorisations: $168 = 2^3 \cdot 3 \cdot 7 = 2^3 \cdot 3^1 \cdot 7^1$

$192 = 2^6 \cdot 3 = 2^6 \cdot 3^1 \cdot 7^0$

- **Greatest common divisor (GCD)**:

$$\begin{aligned}\textbf{gcd}(168, 192) &= \textbf{gcd}(2^3 \cdot 3 \cdot 7\,,\, 2^6 \cdot 3) \\ &= 2^{min(3,6)} \cdot 3^{min(1,1)} \cdot 7^{min(1,0)} = 2^3 \cdot 3^1 \cdot 7^0 = 24\end{aligned}$$

- **Least common multiple (LCM)**:

$$\begin{aligned}\textbf{lcm}(168, 192) &= \textbf{lcm}(2^3 \cdot 3 \cdot 7\,,\, 2^6 \cdot 3) \\ &= 2^{max(3,6)} \cdot 3^{max(1,1)} \cdot 7^{max(1,0)} = 2^6 \cdot 3^1 \cdot 7^1 = 1344\end{aligned}$$

- We have: $\gcd(168, 192) \cdot \textbf{lcm}(168, 192) = 24 \cdot 1344 = 32,256.$

Representation of Integers

- Let $b > 1$ be some integer, which we call the **base**.
  Every positive integer n can be written in the **unique** form:

$$n = d_k\, b^k + d_{k-1}\, b^{k-1} + d_1\, b^1 + d_0$$

where $0 \leq d_i < b$, $i = 0, 1, \ldots, k$ and $d_k \neq 0$.

Representation of Integers

- Let $b > 1$ be some integer, which we call the **base**.
  Every positive integer n can be written in the **unique** form:

$$n = d_k\, b^k + d_{k-1}\, b^{k-1} + d_1\, b^1 + d_0$$

  where $0 \leq d_i < b$, $i = 0, 1, \ldots, k$ and $d_k \neq 0$.
- The sequence $d_k d_{k-1} \ldots d_1 d_0$ is called the **base** b **expansion of** n.
  - $b = 10$: decimal expansion. $\Sigma = \{0, 1, 2, 3 \ldots, 9\}$
  - $b = 2$: binary expansion. $\Sigma = \{0, 1\}$
  - $b = 8$: octal expansion. $\Sigma = \{0, 1, 2, 3 \ldots, 7\}$
  - $b = 16$: hexadecimal expansion.
    $\Sigma = \{0, 1, 2, 3 \ldots, 9, A, B, C, D, E, F\}$

Outline

1 Intro

2 Sets and Subsets

3 Operations on Sets

4 Sequences

5 String and Languages

6 Integers

7 Matrices

8 Mathematical Structures

Definition of a Matrix

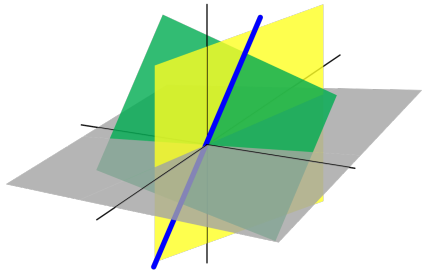
- A **matrix** is a rectangular array of numbers, arranged in (horizontal) rows and (vertical) columns. For instance:

$$A = [a_{ij}] = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

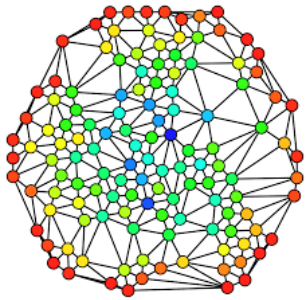
- m (rows) and n (columns) are called the **dimensions** of the matrix.
  We say that the matrix is m**-by-**n, written as $m \times n$.
- If $m = n$, we say that the matrix is a **square matrix of order** m.

Applications of Matrices

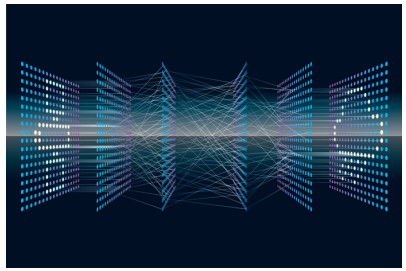
- Linear maps on vector spaces, solving linear systems of equations (Linear Algebra, IIb)



- Graph theory (later in this course)



- Deep learning in AI (image processing, natural language processing)



And much more...

Special Matrices

Square matrices (n rows, n columns).

The identity matrix:

$$I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}$$

The zero matrix Z_n:

$$Z_n = \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}$$

A diagonal matrix:

$$D = \begin{pmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix}$$

Addition of Matrices

- Let A and B be two matrices of the same dimensions.
- Their **sum** $A + B = C$ is the matrix with entries $c_{ij} = a_{ij} + b_{ij}$.
- For example, when A and B have dimension 3×2:

$$A + B = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{pmatrix}$$

$$= \begin{pmatrix} a_{11}+b_{11} & a_{12}+b_{12} \\ a_{21}+b_{21} & a_{22}+b_{22} \\ a_{31}+b_{31} & a_{32}+b_{32} \end{pmatrix}$$

Vectors and Scalar Product

Vectors:

- A **row vector** of length n is a $1 \times n$ matrix.
- A **column vector** of length n is a $n \times 1$ matrix.

Vectors and Scalar Product

Vectors:

- A **row vector** of length n is a $1 \times n$ matrix.
- A **column vector** of length n is a $n \times 1$ matrix.

Suppose we have a row vector and a column vector of the same length, say n. Their **scalar product** is the number (scalar) obtained as follows:

$$\begin{pmatrix} a_{i1} & a_{i2} & \cdots & a_{in} \end{pmatrix} \cdot \begin{pmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{nj} \end{pmatrix} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

Product of Matrices (1)

- Let A be an $m \times k$ matrix and B an $k \times n$ matrix.
- Their **product** $AB = C = [c_{ij}]$ is the $m \times n$ matrix with entries

$$c_{ij} = \text{scalar product of row } i \text{ of } A \text{ and column } j \text{ of } B.$$

Product of Matrices (1)

- Let A be an $m \times k$ matrix and B an $k \times n$ matrix.
- Their **product** $AB = C = [c_{ij}]$ is the $m \times n$ matrix with entries

  c_{ij} = scalar product of row i of A and column j of B.
- An example ($m = 2, k = 3, n = 2$):

$$A = \begin{pmatrix} 2 & 3 & -4 \\ 1 & 2 & 5 \end{pmatrix} \quad B = \begin{pmatrix} 7 & 1 \\ -6 & 6 \\ 9 & 0 \end{pmatrix} \quad AB = [c_{ij}] = \begin{pmatrix} -40 & 20 \\ 40 & 13 \end{pmatrix}$$

since

$$\begin{aligned} c_{11} &= (2 \cdot 7) + (3 \cdot -6) + (-4 \cdot 9) &&= -40 \\ c_{12} &= (2 \cdot 1) + (3 \cdot 6) + (-4 \cdot 0) &&= 20 \\ c_{21} &= (1 \cdot 7) + (2 \cdot -6) + (5 \cdot 9) &&= 40 \\ c_{22} &= (1 \cdot 1) + (2 \cdot 6) + (5 \cdot 0) &&= 13 \end{aligned}$$

Product of Matrices (2)

More generally, if A is a 2×3 matrix and B is a 3×2 matrix:

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} \qquad B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{pmatrix}$$

then their product AB is the 2×2 matrix C defined as follows:

$$\begin{pmatrix} c_{11} = a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} & c_{12} = a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} \\ c_{21} = a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} & c_{22} = a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} \end{pmatrix}$$

- The matrix product is **associative**, i.e., $A(BC) = (AB)C$
- In general, the product is **not commutative**, i.e., $AB \neq BA$. (Why?)

Transpose of a Matrix

- The **transpose** of an $m \times n$ matrix A is the $n \times m$ matrix obtained by interchanging the rows and columns of A.
- Example:

$$A = \begin{pmatrix} 2 & 3 & -4 \\ 1 & 2 & 5 \end{pmatrix} \quad A^T = \begin{pmatrix} 2 & 1 \\ 3 & 2 \\ -4 & 5 \end{pmatrix}$$

- More generally,

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \quad A^T = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \vdots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{pmatrix}$$

- A matrix is called **symmetric** if $A = A^T$.

Boolean Operations as Bit Operations

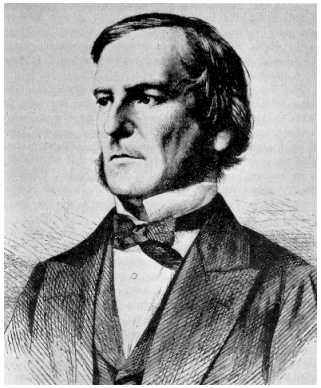


Figure: Mathematician George Boole (1815-1864)

- Boolean values as bits: 1 ("true", $\top$) or 0 ("false", $\bot$).
- Boolean operations on bits $a, b \in \{0, 1\}$:

$$\begin{array}{lllll} \textbf{Join/OR}: & a \vee b & = & \max\{a, b\} & \\ & a_1 \vee \ldots \vee a_n & = & \max\{a_1, \ldots, a_n\} & \\ \textbf{Meet/AND}: & a \wedge b & = & \min\{a, b\} & = & a \cdot b \\ & a_1 \wedge \ldots \wedge a_n & = & \min\{a_1, \ldots, a_n\} & = & a_1 \cdot \ldots \cdot a_n \end{array}$$

Boolean Matrices, Join and Meet

- A Boolean matrix is a matrix with values/entries 0 or 1.
- Let $\mathbf{A} = [a_{ij}]$ and $\mathbf{B} = [b_{ij}]$ be m-by-n Boolean matrices. We define:

Boolean Matrices, Join and Meet

- A Boolean matrix is a matrix with values/entries 0 or 1.
- Let $\mathbf{A} = [a_{ij}]$ and $\mathbf{B} = [b_{ij}]$ be m-by-n Boolean matrices. We define:

$$\textbf{Join}: \quad \mathbf{A} \vee \mathbf{B} = \mathbf{C} = [c_{ij}] \quad \text{where} \quad c_{ij} = a_{ij} \vee b_{ij}$$

$$\textbf{Meet}: \quad \mathbf{A} \wedge \mathbf{B} = \mathbf{C} = [c_{ij}] \quad \text{where} \quad c_{ij} = a_{ij} \wedge b_{ij}$$

(We say that Join and Meet are defined element-wise.)

Boolean Matrix Product

- Similar to ordinary matrix product, but instead of $+$ and $\cdot$, we use $\vee$ and $\wedge$ (i.e. $\max$ and $\min$, or equivalently, $\max$ and $\cdot$).
- Let $\mathbf{A}$ be an $m \times k$ Boolean matrix, and $\mathbf{B}$ a $k \times n$ Boolean matrix. Their Boolean matrix product $\mathbf{A} \odot \mathbf{B} = \mathbf{C} = [c_{ij}]$ has entries

$$c_{ij} = (a_{i1} \wedge b_{1j}) \vee \ldots \vee (a_{ik} \wedge b_{kj}) = \max\{a_{i1}b_{1j}, \ldots, a_{ik}b_{kj}\}$$

- A particular instance:

$$\mathbf{A} \odot \mathbf{B} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} \odot \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{pmatrix}$$

$$= \begin{pmatrix} \max(a_{11}b_{11}, a_{12}b_{21}, a_{13}b_{31}) & \max(a_{11}b_{12}, a_{12}b_{22}, a_{13}b_{32}) \\ \max(a_{21}b_{11}, a_{22}b_{21}, a_{23}b_{31}) & \max(a_{21}b_{12}, a_{22}b_{22}, a_{23}b_{32}) \end{pmatrix}$$

Outline

1 Intro

2 Sets and Subsets

3 Operations on Sets

4 Sequences

5 String and Languages

6 Integers

7 Matrices

8 Mathematical Structures

Mathematical Structures

We have seen several examples of collections of objects (often a set) with operations on them.

- Examples:

$(\mathbb{R}, +, \times)$	the real numbers with the operations addition and multiplication.
$(\{0,1\}, \vee, \wedge)$	the set of bits with the operations join and meet
$(\mathcal{P}(U), \cup, \cap, ^-)$	the power set of U with the operations union, intersection, and complement
$(Mat_n, +, \cdot, ^T)$	all $n \times n$ matrices with matrix sum, matrix product and transpose.

Mathematical Structures

We have seen several examples of collections of objects (often a set) with operations on them.

- Examples:

$(\mathbb{R}, +, \times)$	the real numbers with the operations addition and multiplication.
$(\{0,1\}, \vee, \wedge)$	the set of bits with the operations join and meet
$(\mathcal{P}(U), \cup, \cap, \overline{\ })$	the power set of U with the operations union, intersection, and complement
$(Mat_n, +, \cdot, ^T)$	all $n \times n$ matrices with matrix sum, matrix product and transpose.

Definition. A mathematical structure is a collection of objects with operations defined on them (and their associated properties).

Arities and Prefix/Infix/Postfix Notation

Arity of an operation:

- Unary: operation takes one argument.
  Example: Complement of a set $\overline{A}$
- Binary: operation takes two arguments.
  Example: Sum of two integers $x + y$
- more generally, n-ary: operation takes n arguments
- nullary ($n = 0$): operation takes no arguments (aka constant).
  Example: the natural number 1.

Arities and Prefix/Infix/Postfix Notation

Arity of an operation:

- Unary: operation takes one argument.
  Example: Complement of a set $\overline{A}$
- Binary: operation takes two arguments.
  Example: Sum of two integers $x + y$
- more generally, n-ary: operation takes n arguments
- nullary ($n = 0$): operation takes no arguments (aka constant).
  Example: the natural number 1.

Notation: Example: union of two sets A and B.

- prefix notation: $\cup(A, B)$
- infix notation: $A \cup B$ (typical for binary operations)
- postfix notation: $(A, B)\cup$

Closure

Definition

Given an n-ary operation $\square$ that takes arguments from a set V, we say that V **is closed with respect to** $\square$ if for all $v_1, \ldots, v_n \in V$, we have $\square(v_1, \ldots, v_n) \in V$.

Examples and non-Examples:

- The set Σ^* of strings is closed w.r.t. concatenation.

Closure

Definition

Given an n-ary operation $\square$ that takes arguments from a set V, we say that V **is closed with respect to** $\square$ if for all $v_1, \ldots, v_n \in V$, we have $\square(v_1, \ldots, v_n) \in V$.

Examples and non-Examples:

- The set Σ^* of strings is closed w.r.t. concatenation.
- The set of natural numbers is closed w.r.t. addition.

Closure

Definition

Given an n-ary operation $\square$ that takes arguments from a set V, we say that V **is closed with respect to** $\square$ if for all $v_1, \ldots, v_n \in V$, we have $\square(v_1, \ldots, v_n) \in V$.

Examples and non-Examples:

- The set Σ^* of strings is closed w.r.t. concatenation.
- The set of natural numbers is closed w.r.t. addition.
- The set of odd natural numbers is not closed w.r.t. addition.

Closure

Definition

Given an n-ary operation $\square$ that takes arguments from a set V, we say that V **is closed with respect to** $\square$ if for all $v_1, \ldots, v_n \in V$, we have $\square(v_1, \ldots, v_n) \in V$.

Examples and non-Examples:

- The set Σ^* of strings is closed w.r.t. concatenation.
- The set of natural numbers is closed w.r.t. addition.
- The set of odd natural numbers is not closed w.r.t. addition.
- The set of all square matrices is closed w.r.t. addition, product and transpose.

Closure

Definition

Given an n-ary operation $\square$ that takes arguments from a set V, we say that V **is closed with respect to** $\square$ if for all $v_1, \ldots, v_n \in V$, we have $\square(v_1, \ldots, v_n) \in V$.

Examples and non-Examples:

- The set Σ^* of strings is closed w.r.t. concatenation.
- The set of natural numbers is closed w.r.t. addition.
- The set of odd natural numbers is not closed w.r.t. addition.
- The set of all square matrices is closed w.r.t. addition, product and transpose.
- The set of 2×3 matrices is closed w.r.t. addition, but not w.r.t. product and transpose.

Properties of Operations (1)

We can think of operation such as $+, \cdot, \vee, \wedge$ etc. in terms of their abstract properties.

- A binary operation $\square$ in an MS $(V, \circ, \square, \ldots)$ is called
  - **associative** if, for all $x, y, z \in V$,

$$(x \square y) \square z = x \square (y \square z)$$

  - **commutative** if, for all $x, y \in V$,

$$x \square y = y \square x$$

  - **idempotent** if, for all $x \in V$,

$$x \square x = x$$

Properties of Operations (1)

We can think of operation such as $+, \cdot, \vee, \wedge$ etc. in terms of their abstract properties.

- A binary operation $\square$ in an MS $(V, \circ, \square, \ldots)$ is called
  - **associative** if, for all $x, y, z \in V$,
$$(x \square y) \square z = x \square (y \square z)$$
  - **commutative** if, for all $x, y \in V$,
$$x \square y = y \square x$$
  - **idempotent** if, for all $x \in V$,
$$x \square x = x$$
- A unary operation $\circ$ in an MS $(V, \circ, \square, \ldots)$ is called **idempotent** if
$$\text{for all } x \in V: \quad \circ(\circ x) = \circ x$$

Properties of Operations (2)

Let $(V, ^\circ, \square, \triangledown)$ be an MS with binary operations $\square$ and $\triangledown$ and unary operation $^\circ$.

- We say that $\square$ **distributes over** $\triangledown$ if, for all $x, y, z \in V$,

$$x \square (y \triangledown z) = (x \square y) \triangledown (x \square z)$$

- The operations satisfy **De Morgan's laws** if, for $x, y \in V$,

$$(x \square y)^\circ = x^\circ \triangledown y^\circ$$
$$(x \triangledown y)^\circ = x^\circ \square y^\circ$$

Identity Elements

Let $\square$ be a binary operation on a set V.
An element e is called an **identity element** of $\square$ if, for all $x \in V$,

$$x \square e = e \square x = x$$

Identity Elements

Let $\square$ be a binary operation on a set V.
An element e is called an **identity element** of $\square$ if, for all $x \in V$,

$$x \square e = e \square x = x$$

Examples:

- $(\mathbb{N}, +)$: 0 is an identity for $+$.
- $(\mathbb{R}, \times)$: 1 is an identity for $\times$.
- $(\mathcal{P}(A), \cup, \cap)$: A is an identity for $\cap$, and $\emptyset$ is an identity for $\cup$.
- $(Mat_n, \cdot)$: the identity matrix I_n is an identity for matrix product.

Identity Elements

Let $\square$ be a binary operation on a set V.
An element e is called an **identity element** of $\square$ if, for all $x \in V$,

$$x \square e = e \square x = x$$

Examples:

- $(\mathbb{N}, +)$: 0 is an identity for $+$.
- $(\mathbb{R}, \times)$: 1 is an identity for $\times$.
- $(\mathcal{P}(A), \cup, \cap)$: A is an identity for $\cap$, and $\emptyset$ is an identity for $\cup$.
- $(Mat_n, \cdot)$: the identity matrix I_n is an identity for matrix product.

Theorem Identity elements are unique.

Identity Elements

Let $\square$ be a binary operation on a set V.
An element e is called an **identity element** of $\square$ if, for all $x \in V$,

$$x \square e = e \square x = x$$

Examples:

- $(\mathbb{N}, +)$: 0 is an identity for $+$.
- $(\mathbb{R}, \times)$: 1 is an identity for $\times$.
- $(\mathcal{P}(A), \cup, \cap)$: A is an identity for $\cap$, and $\emptyset$ is an identity for $\cup$.
- $(Mat_n, \cdot)$: the identity matrix I_n is an identity for matrix product.

Theorem Identity elements are unique.
Proof (the power of abstraction) Suppose e_1 and e_2 are both identity elements for $\square$.

Identity Elements

Let $\square$ be a binary operation on a set V.
An element e is called an **identity element** of $\square$ if, for all $x \in V$,

$$x \square e = e \square x = x$$

Examples:

- $(\mathbb{N}, +)$: 0 is an identity for $+$.
- $(\mathbb{R}, \times)$: 1 is an identity for $\times$.
- $(\mathcal{P}(A), \cup, \cap)$: A is an identity for $\cap$, and $\emptyset$ is an identity for $\cup$.
- $(Mat_n, \cdot)$: the identity matrix I_n is an identity for matrix product.

Theorem Identity elements are unique.
Proof (the power of abstraction) Suppose e_1 and e_2 are both identity elements for $\square$. Then $e_1 \square e_2 = e_1$ because e_2 is an identity. Also $e_1 \square e_2 = e_2$ because e_1 is an identity. Hence $e_1 = e_2$.

Wrapping up

Learning goals: At the end of this week, you are able to

- specify and reason about sets, sequences, integers, and matrices.
- apply basic operations on sets, sequences, integers and matrices.
- give simple mathematical proofs of basic facts about sets, sequences, integers and matrices.
- apply the concept of mathematical structure.

This week:

- Readings and exercises this week: see Brightspace (Week 1).

Remember to go through all information in Brightspace!

End of Part 1